

Day 20 - Lecture 20.16: Variance & Standard Deviation

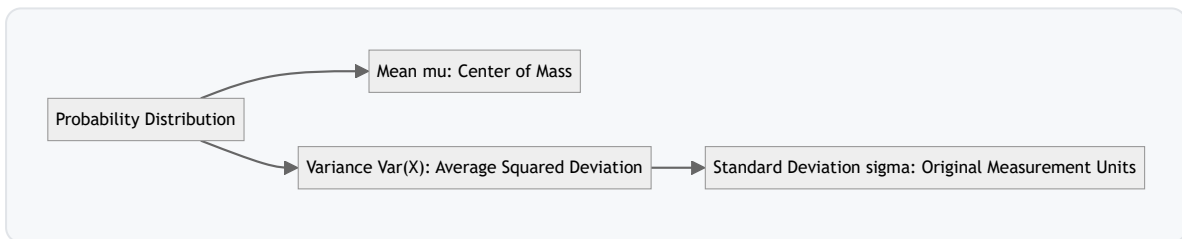
 Python 3.10+

PDF notes.pdf

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1. Overview & Measures of Dispersion

While central tendency locates the center of a probability distribution, **Variance** and **Standard Deviation** quantify its **spread**, dispersion, and uncertainty around that center.



2. Mathematical Formalism

Variance ($\text{Var}(X)$ or σ^2):

The expected squared deviation from the mean $\mu = \mathbb{E}[X]$:

$$\text{Var}(X) = \sigma^2 = \mathbb{E}[(X - \mu)^2]$$

Computational Shortcut Theorem:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Standard Deviation (σ):

The positive square root of variance, returning the dispersion metric back to the original physical units of the measurement:

$$\sigma = \sqrt{\text{Var}(X)}$$

Algebraic Scaling Properties:

For constants $a, b \in \mathbb{R}$:

$$\text{Var}(aX + b) = a^2 \text{Var}(X) \quad \text{and} \quad \text{SD}(aX + b) = |a| \text{SD}(X)$$

Variance of Sums:

For arbitrary random variables X and Y :

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

If X and Y are statistically independent, then $\text{Cov}(X, Y) = 0$:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

3. Python Implementation

```
import numpy as np

# Sample data
x = np.array([10, 12, 14, 16, 18, 20])
mu = np.mean(x)

# Method 1: Direct definition
var_def = np.mean((x - mu) ** 2)

# Method 2: Computational shortcut  $E[X^2] - (E[X])^2$ 
var_shortcut = np.mean(x ** 2) - (mu ** 2)

# Standard deviation
std_dev = np.sqrt(var_def)

print(f"Mean mu: {mu:.2f}")
print(f"Variance (Definition): {var_def:.4f}")
print(f"Variance (Shortcut): {var_shortcut:.4f}")
print(f"Standard Deviation sigma: {std_dev:.4f}")
print(f"Numpy population std check: {np.isclose(std_dev, np.std(x))}")
```

4. Key Takeaways & Interview Points

- **Units:** Variance is measured in squared units (units^2); Standard Deviation is measured in the original linear units.
- **Additive Shifts do NOT affect Variance:** Adding a constant b shifts the entire distribution without spreading it ($\text{Var}(X + b) = \text{Var}(X)$).
- **Multiplicative Scaling:** Multiplying by a amplifies variance quadratically (a^2).